CONSUMER LOAN RISK & CREDIT PERFORMANCE ANALYSIS

Python-Based Data Understanding, Cleaning, EDA and Visualization

PROJECT DOCUMENTATION

Business question: What borrower and loan characteristics are associated with higher credit risk, loan performance, and repayment problems?

1. Executive Summary

This project examines a consumer-loan dataset to understand portfolio composition, loan characteristics, credit grades, and repayment status. The workflow covers data understanding, targeted cleaning, exploratory data analysis (EDA), visualizations, and a consolidated Python script. The analysis is descriptive: it identifies patterns and comparisons in the available data, but does not establish causation or predict future defaults.

The original dataset contains 10,000 rows and 56 columns. After removing one unnecessary index-like column, the cleaned dataset contains 10,000 rows and 55 columns. Missing values in joint-application fields are retained where they may indicate that those fields are not applicable to individual applications.

2. Project Overview

Item	Description
Project	Consumer Loan Risk & Credit Performance Analysis
Domain	Consumer lending / BFSI
Business question	What borrower and loan characteristics are associated with higher credit risk, loan performance, and repayment problems?
Tools	Python, pandas, NumPy, Matplotlib, Jupyter Notebook, VS Code
Scope	Data understanding, cleaning, EDA, visualization, reusable scripting
Out of scope	Machine learning, predictive default modelling, and Power BI in the current scope

3. Business Question and Objectives

The project investigates whether borrower and loan characteristics show meaningful differences across loan grades, purposes, and loan statuses.

- Understand the structure and quality of the dataset.
- Clean selected fields while preserving meaningful missing values.
- Explore loan status, purpose, loan amount, interest rate, debt-to-income ratio, grade, and homeownership.
- Use visualizations to communicate selected portfolio patterns.
- Document evidence, implications, and limitations without overstating conclusions.

4. Dataset Description

Dataset stage	Rows	Columns	Notes
Original	10,000	56	Raw source file; preserve unchanged.
Cleaned	10,000	55	Unnecessary Unnamed: 0 column removed.

The dataset contains borrower attributes, credit-history fields, loan terms and amounts, grades, loan statuses, and repayment-related measures. Exact field meanings should be verified against the source data dictionary, if available.

5. Tools and Technologies

- Python — core programming language for analysis.
- pandas — data loading, cleaning, grouping, and summary statistics.
- NumPy — numerical computing support.
- Matplotlib — static charts.
- Jupyter Notebook — stepwise exploration and documentation of analytical decisions.
- VS Code — consolidated, reusable Python script.

6. Methodology and Workflow

1. Data understanding: inspect columns, dimensions, data types, missing values, duplicates, categorical counts, and descriptive statistics.
2. Data cleaning: remove the unnecessary index-like column, apply targeted missing-value handling, standardize selected text, and convert issue_month to datetime.
3. EDA: compare loan status, purpose, amounts, interest rates, debt-to-income ratios, grades, and homeownership.
4. Visualization: create four Matplotlib charts to communicate selected distributions and comparisons.
5. Scripting: consolidate the workflow into one Python script for repeatable execution.

7. Data Understanding

Initial inspection covered column names, dataset shape, data types, first rows, missing-value counts, duplicate rows, categorical distributions, descriptive statistics, and loan-status percentages. This established a baseline before transformations.

8. Data Cleaning Decisions and Justifications

Field / operation	Action	Reason / caution
Unnamed: 0	Removed	Unnecessary index-like column.
emp_title	Fill missing with Unknown; trim whitespace	Makes missing titles explicit and standardizes text.
emp_length	Fill missing values with median	Targeted numeric imputation; introduces estimated values.
num_accounts_120d_past_due	Fill missing with 0	Assumes missing means zero; verify source semantics.
debt_to_income	Fill missing values with median	Targeted imputation; may influence group averages.
Delinquency / inquiry recency fields	Fill missing values with 0	Only valid if source definitions support interpreting missing as none.
issue_month	Convert to datetime using %b-%y	Supports date-aware handling.
Joint-application fields	Retain missing values	May be not applicable to individual applications; avoid inventing values.
Duplicates	Checked after cleaning; 0 found	No exact duplicate rows detected.

Cleaning caution: imputing a missing recency or delinquency value with zero is semantically appropriate only if the dataset documentation supports that interpretation. Describe these choices transparently.

9. Exploratory Data Analysis

The EDA examined:

- Loan status and loan purpose frequency distributions.
- Loan amount summary statistics and average amounts by purpose and status.
- Interest-rate summary statistics and average interest rate by grade.
- Average debt-to-income ratio by loan status.
- Loan status distributions across grades and homeownership categories.

10. Key Findings and Supporting Evidence

Finding	Evidence	Potential business implication	Limitation
Portfolio is dominated by Current loans.	9,375 of 10,000 loans are Current (93.75%).	Current status represents most records at the observed snapshot.	Current does not guarantee eventual repayment; loan age and timing matter.
Debt consolidation is the most common purpose.	5,144 loans.	This purpose is a major part of the observed portfolio.	Frequency alone does not establish profitability or risk.
Interest rates generally rise from Grade A toward Grade G.	Grouped mean rates showed an upward pattern; verify exact values from notebook.	Consistent with grade-level pricing differentiation.	Descriptive association does not establish the cause.
Average loan amount varies by purpose and status.	Group means were calculated in EDA.	May help investigate lending exposure across segments.	Copy exact values from executed notebook output before publication.
Average DTI differs by loan status.	Group means were calculated in EDA.	DTI may warrant further investigation.	Averages alone do not establish predictive power or causation.

Rare-status caution: only 7 loans are classified as Charged Off in the supplied counts. This is too small a sample for strong generalizations. Show sample sizes when discussing rare statuses.

11. Visualizations

Chart	Purpose
Loan Status Distribution	Shows the number of loans in each status category.
Loan Purpose Distribution	Shows which purposes occur most frequently.
Average Interest Rate by Loan Grade	Compares average interest rates across grades.
Average Loan Amount by Purpose	Compares average loan amounts across purposes.

Interpret charts alongside category counts and data limitations, especially where sample sizes are small.

12. Business Implications

- Portfolio composition can be summarized through loan-status and loan-purpose distributions.
- Grade-level interest-rate differences can be reviewed as evidence of pricing differentiation, without assuming causation.

- Differences in loan amounts and debt-to-income ratios can guide further investigation, but do not prove default risk.
- Rare outcomes require cautious interpretation and additional evidence before operational decisions.

13. Limitations and Analytical Cautions

- This is descriptive analysis; it does not predict defaults or establish causal relationships.
- Only seven Charged Off records appear in the supplied status counts, limiting analysis of that outcome.
- Imputation decisions can influence statistics and should be documented.
- Joint-application fields retain missing values because they may not apply to individual applications.
- Dataset time coverage, sampling method, and field definitions should be verified before making broader claims.
- Do not label the Current-loan share as a default rate or as proof of low risk.

14. Conclusion

The project establishes a reproducible descriptive workflow for understanding a consumer-loan portfolio. It summarizes the dominant loan status and purpose, examines differences in loan amounts and interest rates, and compares debt-to-income ratios across statuses. The results provide a basis for further investigation, but the small number of Charged Off records and descriptive nature of the analysis limit strong risk conclusions. Finalize exact summary statistics from executed notebook outputs.